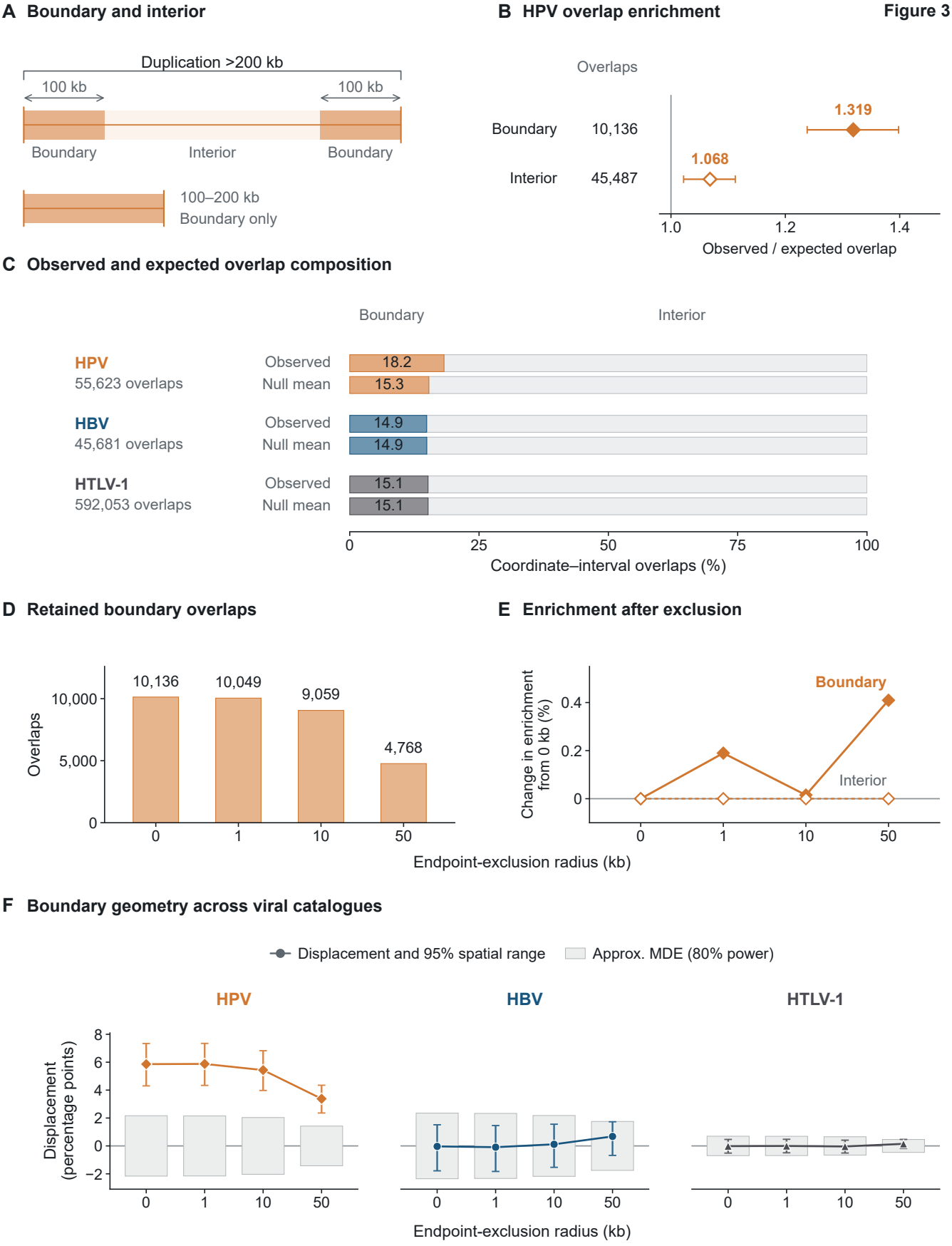

A Boundary and interior
Duplication >200 kb
100 kb
100 kb
Boundary
Interior
Boundary
100–200 kb
Boundary only
B HPV overlap enrichment
Overlaps
Boundary 10,136
Interior 45,487
1.319
1.068
1.0
1.2
1.4
Observed / expected overlap
C Observed and expected overlap composition
Boundary
Interior
HPV
55,623 overlaps
Observed 18.2
Null mean 15.3
HBV
45,681 overlaps
Observed 14.9
Null mean 14.9
HTLV-1
592,053 overlaps
Observed 15.1
Null mean 15.1
0 25 50 75 100
Coordinate–interval overlaps (%)
D Retained boundary overlaps
Overlaps
10,000
5,000
0
10,136
10,049
9,059
4,768
0 1 10 50
E Enrichment after exclusion
Change in enrichment from 0 kb (%)
0.4
0.2
0
Boundary
Interior
0 1 10 50
Endpoint-exclusion radius (kb)
F Boundary geometry across viral catalogues
Displacement and 95% spatial range
Approx. MDE (80% power)
HPV
HBV
HTLV-1
Displacement (percentage points)
8 6 4 2 0 −2
0 1 10 50
Endpoint-exclusion radius (kb)